Purpose
To detect and track the players as well as the pickleball across a video. It should note the playing area's key points to determine player's position, measure distances, determine if the ball was in/out-of-bounds, and calculate movement speeds (shot speed, avg shot speed, player speed, avg player speed).

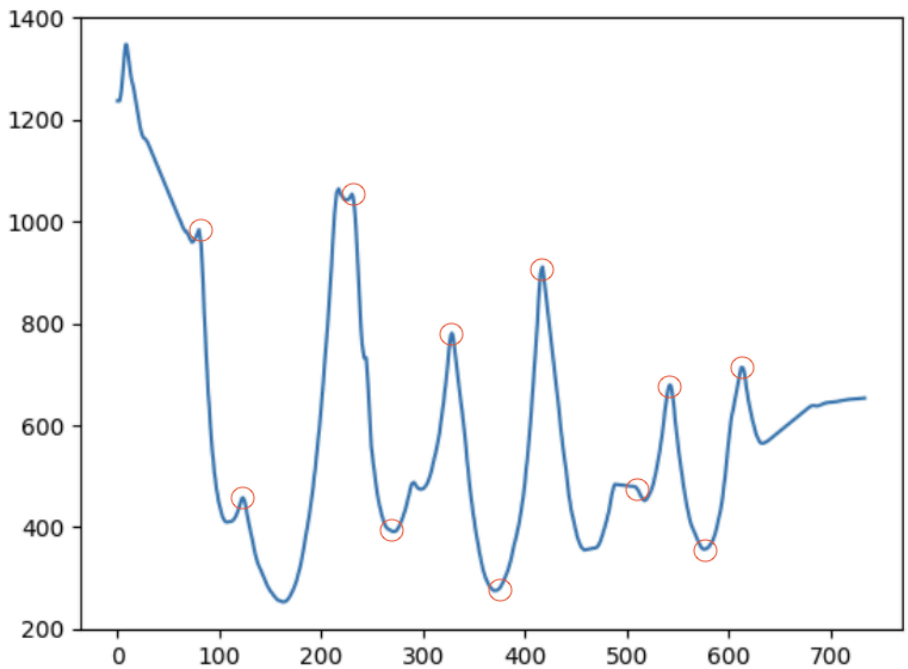
Resources
Code In a Jiffy - YouTube channel with the followed project walkthrough
Yolo - Object detection model used to detect the pickleball and players
Pytorch - Builds models to predict court's key points
Ultralytics - Library that makes YOLO easy to use
Roboflow - Create my dataset to train the model
Google Colab - Runs model training with free cloud GPUs
Visual Studio Code - Edit and runs code locally, working with Python specifically
Keypoint detection - https://youtu.be/aBVGKoNZQUw?si=1aMz7MOHCjd4qi9X and https://youtu.be/uWP6UjDeZvY?si=yuBViZ9CSUwMtQkC
Player/ball tracking and measurements - https://youtu.be/L23oIHZE14w?si=DzItELvZpfFTofJq

Workflow

Problems Encountered

1. Ultralytics was not recognized after installation
    a. Opened Command Palette (Cmd + Shift + P) > Python: Select Interpreter > Enter interpreter path > TYPE "/Library/Frameworks/Python.framework/Versions/3.13/bin/python3"
2. Getting this error when trying to run roboflow + yolo training: RuntimeError: Dataset '/content/Pickleball-Detector-2/data.yaml' error ❌ '/content/Pickleball-Detector-2/data.yaml' does not exist
    a. data.yaml file was not being created in google colab. Had to run the roboflow code in visual studio code and copy the folder with the data.yaml file to google colab
3. Pickleball is being inconsistently tracked in test video
    a. Increased the amount of data in training model
    b. Lowered confidence threshold to accept more bounding box attempts
        i. Found that 0.05 confidence excluded all wrong guesses while including the most attempts possible
    c. Added interpolation to track the ball's path and movement between frames where it is undetected by the model.
4. Unable to get the key points to display on the court for any image test. Predictions using json files have the key points displayed grouped up off court.
    a. Switched to trainYOLO for dataset and yolov8 for training. 100 epochs, 640 image size. → still had inaccuracies
    b. Instead, prompts the user to select the key points of the court based on the first frame of the video.

5. The four closest players chosen would sometimes be non-players
    a. Instead of finding the minimum distance of each track id to any of the court keypoints and choosing the four track ids with the lowest minimum distances, I instead calculated the center of the court and got the four track ids closest to the center.
6. The hits are not being properly tracked, having many false positives and missing hits.



    a.

	x1	y1	x2	y2	mid_y	mid_y_rolling_mean	delta_y	ball_hit	delta_y_smooth
2	561.891785	1209.934448	590.575073	1263.674072	1236.804260	1236.854106	-0.024923	1	NaN
8	554.647888	1332.240479	584.457764	1374.000000	1353.120239	1347.758895	15.208736	1	NaN
67	820.271301	961.802368	846.464783	998.117615	979.959991	979.963757	-2.860840	1	-4.067106
68	820.274902	961.804871	846.466431	998.130493	979.967682	979.965814	0.002057	1	-3.729031
73	850.452698	943.469971	879.058838	981.561646	962.515808	959.551410	-3.490375	1	2.244525
80	870.308228	975.714172	897.064026	1010.945251	993.329712	984.360901	4.406946	1	-11.509906
107	1379.007464	398.902710	1397.320557	421.997585	410.450147	409.725652	-1.005718	1	-1.586954
123	1494.163208	441.098460	1508.164836	462.262909	451.680684	457.851936	1.616070	1	-0.446040
162	1516.321167	241.449890	1530.366699	262.608612	252.029251	253.083749	-0.698848	1	-0.204115
217	1350.111816	1034.096558	1376.911865	1069.710449	1051.903503	1064.321985	2.927728	1	7.655230
225	1337.126465	1028.016479	1364.595215	1061.663574	1044.840027	1042.555774	-0.053271	1	0.646085
230	1331.808960	1044.371094	1356.016235	1077.769043	1061.070068	1053.924072	3.246008	1	-8.227225
270	1443.132446	376.263062	1462.325684	403.499359	389.881210	391.325049	-0.961903	1	-0.136287
291	1487.260742	466.804413	1503.990723	491.288330	479.046371	487.630042	0.577039	1	1.545462
297	1522.148193	461.983266	1541.592977	487.463654	474.723460	474.404253	-0.863346	1	-1.643182
328	1690.265137	744.295127	1712.042425	781.855785	763.075456	781.092074	4.929963	1	3.042739
370	1333.783936	261.515808	1354.531372	287.966583	274.741196	275.440726	-0.344373	1	-0.623186
417	1476.387066	852.136375	1504.274112	906.594826	879.365601	910.948193	4.542468	1	4.499557
458	1526.495768	342.529551	1544.258643	368.981258	355.755405	354.925851	-1.865078	1	-1.469541
488	1503.810397	471.104309	1519.136364	496.095685	483.599997	484.023416	3.975795	1	4.849746
517	1498.071313	442.194635	1513.298486	465.090253	453.642444	452.214411	-1.052375	1	-1.246551
542	1230.238159	669.424438	1246.947021	702.431580	685.928009	679.476886	3.226617	1	0.764698
576	1312.447754	345.645294	1326.785400	367.898346	356.771820	356.613449	-0.086786	1	-0.886375
613	1139.830688	680.549500	1158.390747	709.414062	694.981781	713.622416	0.120907	1	0.007723
633	1280.299194	553.613831	1297.996704	577.304321	565.459076	564.450806	-0.483594	1	-0.746722
683	1470.379956	626.188391	1485.823145	650.023730	638.106061	638.773522	0.105732	1	0.465265
686	1477.456665	625.106995	1493.274048	649.642212	637.374603	637.811329	-0.294731	1	0.276616

The circles are the frames in which the ball was hit while the graph displays the ball’s y position. The following table is all the frames recorded as hits

7. Player positions and pickleball position on mini court are not being properly tracked

a.